

Aditya Kulkarni

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PROFESSIONAL SUMMARY

Builder specializing in AI agents and LLM tooling -- architected an agentic AI orchestration platform handling 2,000 inference requests/second (AlgoPulse), built production multi-model LLM tooling with MCP integrations at Isomer AI, and designed an AI quality eval framework benchmarking Claude, GPT-4o, and Gemini across 2,000 test cases. MS CS from NC State (May 2026), IEEE-published ML work, 6+ hackathon wins. Strong Python and TypeScript; build out of curiosity first, production second.

CORE COMPETENCIES

LLM Agents + MCP

Agentic Workflow Orchestration

Python + TypeScript

Multi-Model AI (Claude/GPT-4o/Gemini)

Foundation Model Evaluation

Kafka + Temporal.io

REST APIs + Docker

0-to-1 Product Development

WORK EXPERIENCE

Isomer AI May 2025 - Present

Product Engineer Intern - Raleigh, NC

- Built production **multi-model LLM platform** integrating Claude, GPT-4o, and Gemini (8+ models) via MCP; implemented Temporal.io agentic workflow orchestration where AI agents operate within human approval loops end-to-end
- Shipped 10+ backend features across **394 commits and 7 production deployments** -- TypeScript, Node.js, Express, PostgreSQL, Redis, Auth0, Docker, GitHub Actions CI/CD

NC State University Apr 2025 - Present

Graduate Research Assistant - Raleigh, NC

- Built GCP Cloud Run platform for 100+ researchers exploring 500GB genomic datasets; improved ML model accuracy by 12% through hyperparameter tuning and time-series analysis

Mphasis (HP Company) Jun 2024 - Aug 2024

Software Developer Intern

- Built 5+ AI-driven automation workflows including health claim automation (OCR+NLP+ML, 25% accuracy improvement); automated AWS and GCP deployments

PROJECTS

AlgoPulse -- Agentic AI Orchestration Platform Agents + Temporal + Kafka github.com/Aditya-k24/AlgoPulse

Architected an event-driven agentic platform where AI agents handle async inference workflows with Temporal.io state management and Kafka fault tolerance at 2,000 requests/second. Identified API-polling bottleneck mid-build and re-architected the entire delivery layer -- demonstrating the design-feedback-rebuild cycle Loop describes for its AI tracks. TypeScript, Node.js, React.

TypeScript, Node.js, React, Supabase Edge Functions, Temporal.io, Kafka

LLM Eval Harness -- Foundation Model Benchmarking Claude + GPT-4o + Gemini github.com/Aditya-k24/llm-eval-harness

Evaluation framework benchmarking Claude Sonnet 4.6, GPT-4o, and Gemini 2.5 Pro across 2,000 frozen test cases. Tracks ExactMatch, TokenF1, grounded hallucination rate, p50/p95 latency, and cost/call; statistical significance via bootstrap CIs and McNemar test. Reusable prompt templates and evaluation infrastructure -- the kind of tooling needed to measure foundation model improvements for a domain-specific model.

Python, Anthropic/OpenAI/Google GenAI SDKs, DuckDB, Parquet, Streamlit, Docker

Glance AI -- Compliance RAG System RAG + Python github.com/Aditya-k24/GlanceAI

Privacy-first RAG system for regulated domains with per-tenant vector isolation preventing context leakage; deterministic output architecture for factual correctness. Built out of curiosity about what trustworthy AI requires beyond prompt engineering.

Python, LLMs, Vector Databases, RAG

Mapify -- IEEE Published ML/NLP Tool IEEE Published github.com/Aditya-k24/mapify

T5-transformer pipeline converting 1,500+ PDFs/images into knowledge graph mind maps; reduced academic review time by 60%. NLTK and NetworkX for structure extraction; presented to 500+ attendees at IEEE conference. Published before LLMs made this easy -- built it because the problem was interesting.

Python, T5 Transformer, NLTK, NetworkX, NLP

EDUCATION

North Carolina State University -- MS Computer Science Aug 2024 - May 2026

Raleigh, NC | Courses: Artificial Intelligence, Operating Systems, Algorithms, Software Engineering, Data Science

Mumbai University -- BTech Computer Science Aug 2020 - May 2024

CGPA: 9.32/10 | Awards: 6+ national hackathon wins including Top 10 of 100+ teams (HackNC24); Google Developer Student Club Lead (500+ students)

SKILLS

AI/LLM: Claude API, OpenAI, Gemini, MCP, LangChain, RAG, PyTorch, Hugging Face, Temporal.io, prompt engineering

Languages: Python, TypeScript, Java, JavaScript, SQL

Backend: Node.js, FastAPI, Express, Kafka, PostgreSQL, Redis, MongoDB, Docker, GitHub Actions

Cloud: AWS, GCP (Cloud Run), Kubernetes, Terraform